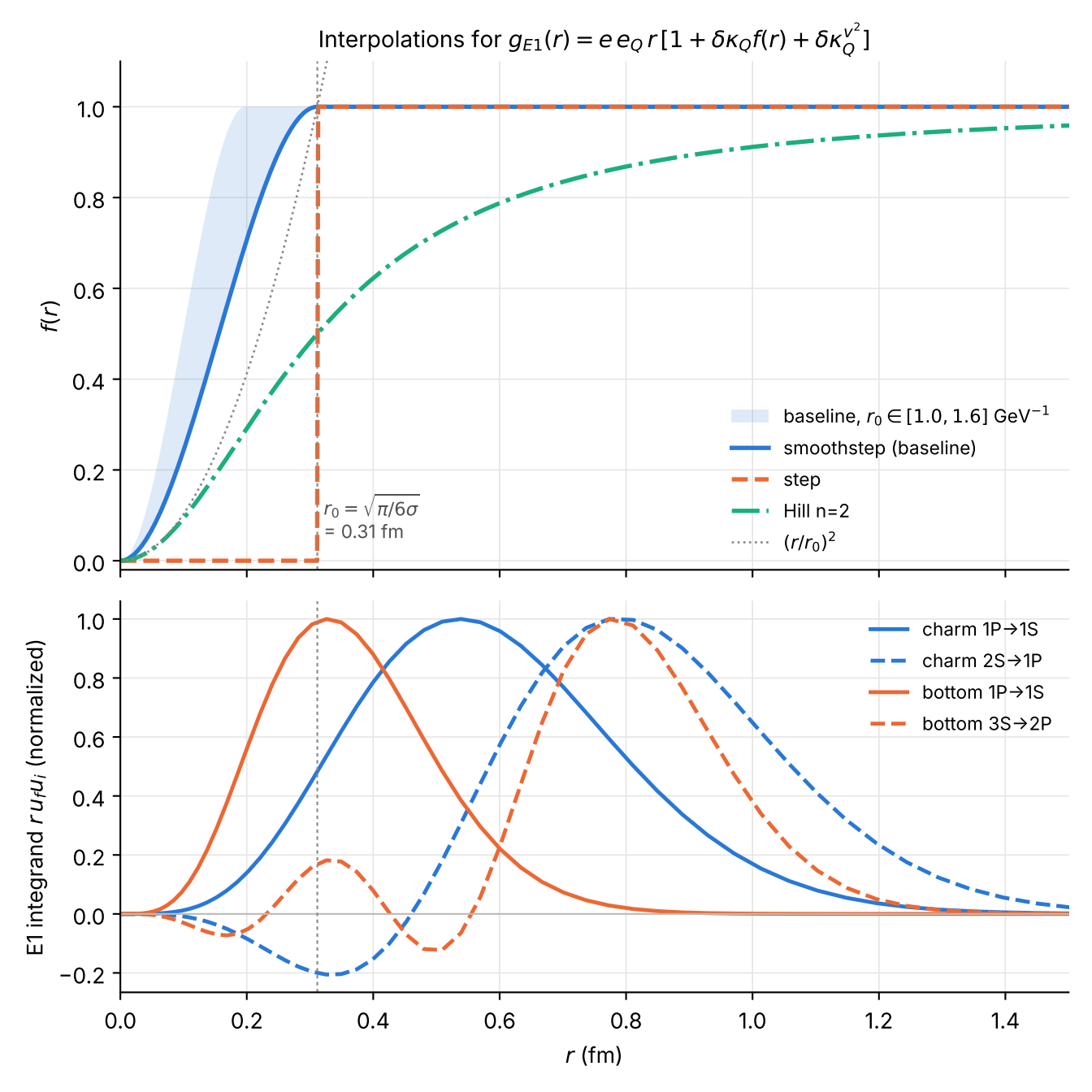
Interpolations for $g_{E1}(r) = e\, e_Q\, r\,[1 + \delta\kappa_Q f(r) + \delta\kappa_Q^{v^2}]$
f(r)
1.0
0.8
0.6
0.4
0.2
0.0
$r_0 = \sqrt{\pi/6\sigma}$
= 0.31 fm
baseline, $r_0 \in [1.0, 1.6]$ GeV^{-1}
smoothstep (baseline)
step
Hill n=2
$(r/r_0)^2$
E1 integrand $r\, u_f u_i$ (normalized)
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
charm 1P→1S
charm 2S→1P
bottom 1P→1S
bottom 3S→2P
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
r (fm)